

Appendix

A.1 – Instrumentation Development of an X-ray Detected Ferromagnetic Resonance Spectrometer

A.1.1 – Introduction

X-ray detected ferromagnetic resonance (XFMR) is a technique based on x-ray magnetic circular dichroism (XMCD) for studying the time-resolved precessional motion of spins undergoing ferromagnetic resonance (FMR) [132–135]. XFMR combines the sensitivity of FMR spectroscopy to the magnetization dynamics of a sample with the chemical sensitivity of x-ray absorption spectroscopy which serves as the underlying detection mechanism. Thus, XFMR is a powerful tool to investigate the magnetization dynamics of complex material systems and enables the element and valence specific detection of magnetization dynamics. Such detection capabilities are also instrumental for investigating the propagation of spin currents in complex heterostructure systems [132].

At the Advanced Light Source beamline (BL) 4.0.2 (henceforth referred to as BL 4.0.2), the XMCD signal measured during FMR (in the following referred to as XFMR signal) is detected through the total optical photoluminescence yield (LY) signal of the sample [132,133,136]. The strength of the LY signal, and thus the XFMR signal, depends on several factors intrinsic to both the sample properties and the instrumentation used. In terms of the samples, the detected LY signal can strongly depend on factors, such as sample temperature and choice of substrate [137]. Generally, the greater the LY efficiency (i.e., conversion of x-ray flux into visible photons) of a substrate, the easier it is to detect and measure a potential XFMR signal. Thus, substrates such as MgO and MgAl₂O₄ (MAO), which exhibit large room temperature LY efficiencies of

approximately 0.1 and 0.03, respectively, are preferred over substrates with small room temperature LY efficiencies, like LSAT (0.004) and STO (0.0009) [137].

Therefore, the selection of a suitable substrate should partially be guided by experimental requirements (i.e., a substrate with greater LY efficiency may be preferable for LY-based XAS studies), however, epitaxial single crystal thin films cannot be grown on arbitrarily chosen substrates: similar crystal structures must be maintained between the film and the substrate. Thus, thin film growth of complex oxides on highly luminescent substrates may not be feasible.

If the LY emission of the sample cannot be further optimized, then an alternative route to enhance the LY signal is to improve the LY detection hardware. In order to enable the investigation of LSMO-based epitaxial heterostructures with XFMR, which are nominally grown on LSAT or STO substrates, I investigated and implemented methods to enhance the detection sensitivity of the XFMR instrumentation at BL 4.0.2. Comparative XAS studies performed between the dedicated XFMR and LY holders revealed the former did not provide sufficient signal sensitivity to measure samples grown on LSAT substrates. By upgrading the photodiode (PD) used in the XFMR holder, I found an improvement in the LY detection capability, where the XFMR holder measures LY signal sensitivity comparable to those produced by the standard LY holder.

A.1.2 – Principles of XFMR

In XFMR, XAS-XMD spectroscopy is used to probe microwave (GHz-frequency) excited magnetization dynamics. XFMR is based on FMR spectroscopy, where magnetic moments are driven into precession about a static magnetizing field $\mathbf{H}_{static}$ through an oscillating RF magnetic field $\mathbf{H}_{RF}$. In these measurements, $\mathbf{H}_{static}$ is perpendicular to $\mathbf{H}_{RF}$ and an FMR spectrum is collected as a function of $\mathbf{H}_{static}$ magnitude and/or $\mathbf{H}_{RF}$ frequency [138,139]. While FMR detects magnetization dynamics through the absorption of microwave power, in XFMR the underlying

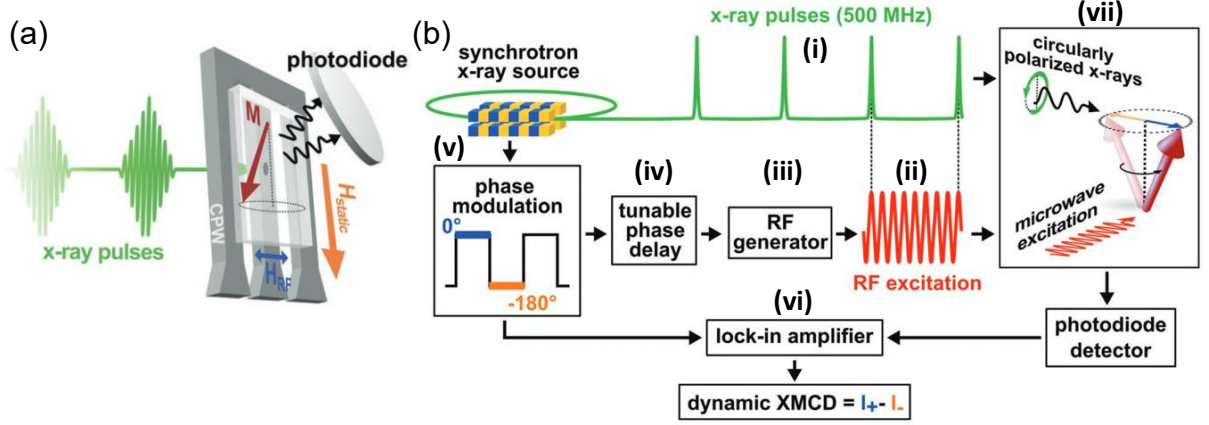


Figure A.1.1: (a) Schematic of the LY signal measurement scheme for XFMR experiments at ALS BL 4.0.2. (b) Schematic of the time-resolved XFMR measurement technique [132]. The role of the instruments and processes labeled with (i-vii) are discussed in the text.

detection mechanism is the XAS-XMCD signal. As discussed later, the XMCD signal can be mapped as a function of time, where changes in the XCMD signal intensity corresponds with the time-varying projection of the magnetic moment $\mathbf{m}$ onto the x-ray beam orientation $\mathbf{k}$ (i.e., $I_{XMCD} \propto \mathbf{m} \cdot \mathbf{k}$).

The precessional behavior of a ferromagnetic system can be described by the Landau-Lifshitz-Gilbert equation of motion

$$\frac{d\mathbf{m}}{dt} = -\frac{\gamma}{1+\alpha^2} (\mathbf{m} \times \mathbf{H}_{eff} + \alpha [\mathbf{m} \times (\mathbf{m} \times \mathbf{H}_{eff})]) \quad \text{Eq. A.1}$$

where $\mathbf{m}$ is the magnetic moment, α is a damping parameter, γ is the gyromagnetic ratio, and $\mathbf{H}_{eff}$ is the effective magnetic field influencing $\mathbf{m}$. $\mathbf{H}_{eff}$ includes both the externally applied field (i.e., $\mathbf{H}_{static}$) and different magnetic effects (e.g., exchange interaction, magnetocrystalline anisotropy, magnetostatics) which can be modeled as effective field terms [62]. Generally, resonant absorption of microwaves by thin film samples depends on several factors including the precessional motion

damping quantified by the Gilbert damping coefficient, α , any sources of magnetic anisotropy, as well as magnetic and structural inhomogeneities [138–140].

In XFMR spectroscopy, time-resolved XMCD is used to directly measure the motion of the precessing magnetic moments. Specifically, the magnetization precesses about $\mathbf{H}_{static}$ oriented perpendicular to the x-ray beam, and the time-dependent projection of the magnetization onto the x-ray beam is probed. Henceforth, any discussion of XFMR instrumentation is made in reference to the experimental setup at BL 4.0.2 [132,133]. In both broadband (i.e., variable radio frequency) FMR and XFMR, the sample is mounted film-side down on top of a coplanar wave guide (CPW) as shown in Figure A.1.1(a). GHz-frequency AC current is sent down the CPW, generating $\mathbf{H}_{RF}$ which oscillates perpendicular to the CPW length. In XFMR, the CPW possesses a $\sim 100\ \mu\text{m}$ diameter hole for the x-rays to pass through and irradiate the sample. The LY detection mode is used to collect the XAS-XMD signal. Time-resolved measurements in XFMR are performed by exploiting the time structure of the ALS synchrotron storage ring, referred to as the master oscillator. The ALS does not produce a continuous beam, but rather behaves as a strobe light producing $\sim 70\ \text{ps}$ long x-ray pulses every $\sim 2\ \text{ns}$. This pulse time corresponds with a master oscillator frequency f_{ALS} of $\sim 500\ \text{MHz}$ ($499.642\ \text{MHz} \pm 6\ \text{kHz}$ [141]).

Figure A.1.1(b) schematically illustrates how the time-resolved XFMR measurement works. The roman numeral indices (i-v) discussed below are in reference to the same indices used in Figure A.1.1(b) to this discussion. XFMR works by phase locking $\mathbf{H}_{RF}$ (ii) with the synchrotron master oscillator (i). The frequency of $\mathbf{H}_{RF}$ is controlled by an RF comb generator (iii) which produces higher harmonics of f_{ALS} between 1 GHz to 4 GHz. Exceeding $\sim 4\ \text{GHz}$ results in a strong suppression of a measurable XFMR signal due to a blurring effect caused by the finite x-ray pulse length at the ALS [132]. Through phase locking the x-ray pulses (i) and RF excitation (ii), a

stroboscopic measurement of the magnetic precession can be performed. If a peak in $\mathbf{H}_{RF}$ is phase locked (i.e., “aligned” in time) with the x-ray pulses, then all subsequent x-ray pulses will arrive at the sample *only* when $\mathbf{H}_{RF}$ is at a peak position. By manually introducing a time offset (i.e., a phase delay (iv)) between the incoming x-ray pulses and $\mathbf{H}_{RF}$, different points in the magnetic precession can be accessed. Thus, the entire time-dependent RF response of a material can be mapped by measuring the XFMR signal as a function of the phase delay time.

As discussed earlier, in XFMR the GHz dynamics is driven by the CPW-generated $\mathbf{H}_{RF}$ while the precessional motion of the driven magnetic moments is measured via XMCD (or XMLD). As a function of time, $I_{XMCD/XMLD}$ changes as the moment precesses. Measurement of this signal is challenging since the cone that the precessing moment traces out has an angle on the order of $\Delta\theta \leq 1^\circ$, where $\Delta\theta$ represents the maximum range of the moment angle variation. Thus, the XFMR signal is expected to be several orders of magnitude smaller than a static XAS-XMCD measurement, where $\Delta\theta = 180^\circ$. For this reason, a lock-in amplifier (LIA, vi) setup is used to extract the small dynamic XMC/LD signal in an X-FMR measurement. A LIA is an instrument capable of detecting and measuring very small AC signals when provided a modulating reference signal for comparison. In an XMCD-XFMR experiment, the reference signal used is a 333 Hz-frequency 180° modulation (v) of the $\mathbf{H}_{RF}$ phase. This modulation allows for the measurement of an XMCD signal which represents the difference between XAS captured with (1) left and right hand polarized light for a fixed $\mathbf{H}$ (e.g., Figure 2.11(b)) or, equivalently, (2) $\mathbf{H}$ applied along two opposite directions for a fixed H magnitude and circular polarization. The phase modulation in XFMR utilizes method (2) to capture the dynamic XMCD signal, where $\mathbf{H}$ corresponds with $\mathbf{H}_{RF}$ which is being “flipped” along two opposite directions through the 180° phase modulation as shown in (vii) in Figure A.1.1.

For the case for a fixed circular polarization, the XFMR signal extracted by the LIA is the dynamic XMCD signal

$$I_{dyn}^{XMCD} = I_+ - I_- \quad \text{Eq. A.1.2}$$

Where I_+ and I_- corresponds with the LY signal intensities at 0° and 180° modulation of $\mathbf{H}_{RF}$, respectively. With respect to Figure 2.11, I_+ and I_- would correspond with the XAS signal produced with RCP and LCP x-rays.

A.1.2 – XFMR Instrumentation at ALS Beamline 4.0.2

Prior to discussing the specific modifications made to the XFMR instrument, it is useful to provide context into the nature of the instrumentation used at BL 4.0.2. BLs are essentially vacuum pipes that the synchrotron-produced x-rays travel down towards the sample contained within experimental stations called “endstations”. BL 4.0.2 contains several endstations, one of which is called the vector magnet (VMM, vector **m**agnetometer) endstation [142]. The VMM consists of an octupole electromagnet which surrounds a small vacuum chamber that the studied sample will reside within. This vacuum chamber is directly connected to the beamline and allows the sample to be exposed to the soft x-rays. The octupole magnet enables the sample to experience up to a 0.5 T magnetic field oriented in any arbitrary direction relative to the x-ray beam.

The samples themselves are attached to the sample holder which consists of a long (~ 1 m) shaft attached to a vacuum flange as schematically shown in Figure A.1.2(a). The length of the sample holder is required to position the sample within the center of the vacuum chamber surrounded by the octupole magnets. Depending on the type of measurement being performed, the copper end piece can either be solid bodies (for electron yield measurements) or be hollowed out to fit a PD (e.g., Figure A.1.2(d)) as in the case of a dedicated LY sample holder, referred to as the “LY holder”. In the case of the “XFMR holder” shown in Figure A.1.2, the sample end piece is

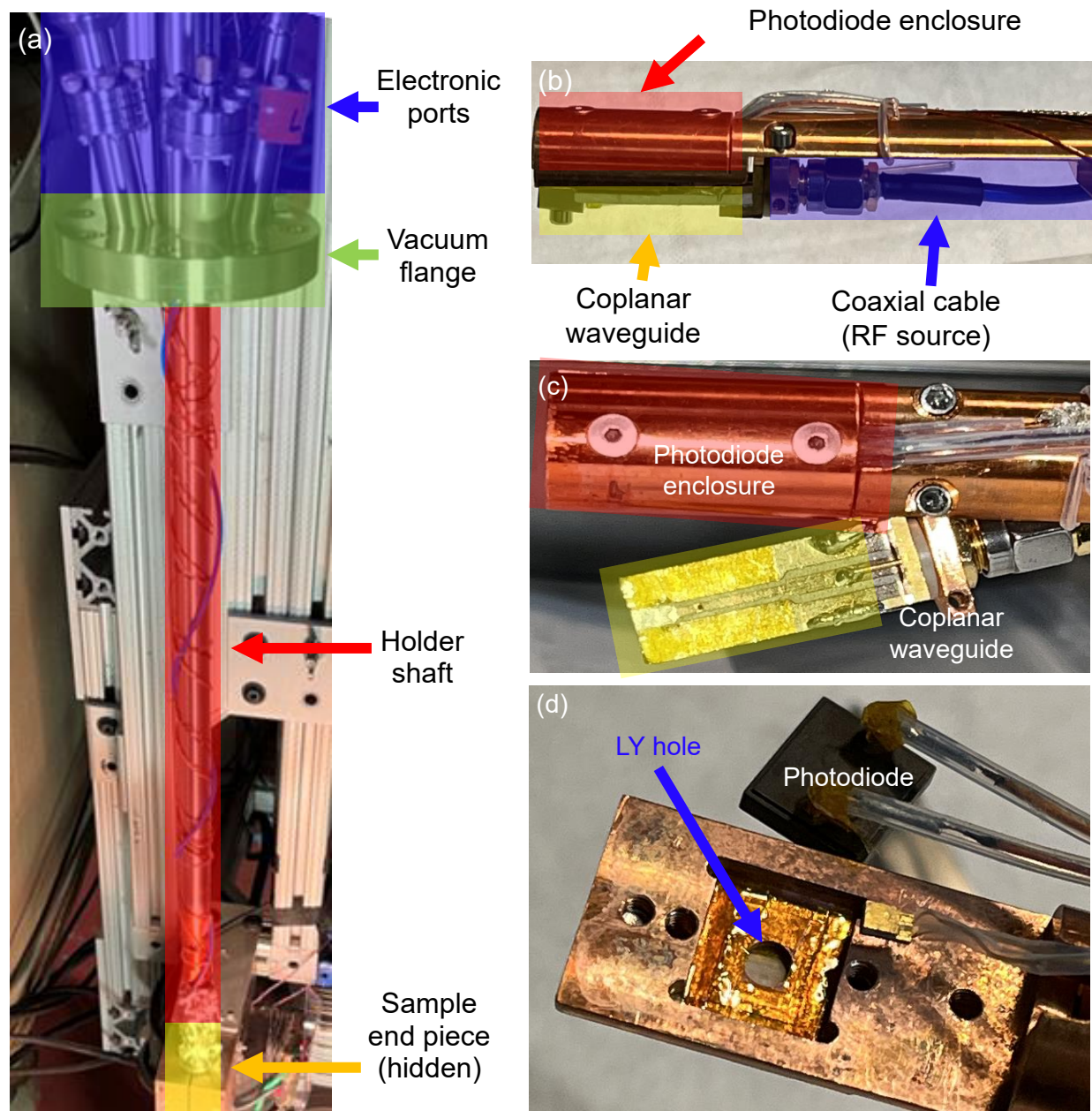


Figure A.1.2: (a) Picture of the XFMR sample holder used in the VMM endstation at BL 4.0.2. Different functional portions of the holder are color coded and labeled. The actual sample end piece in this image is hidden by part of an testing enclosure that the holder is mounted to. (b) Zoom-in side-view image of the XFMR end piece attached to the end of the sample holder (i.e., yellow region in (a)). (c) Back view of (b) with the coplanar wave guide exposed. (d) Back view of (b) with coplanar wave guide mounted and the photodiode enclosure cap and photodiode

comprised of two halves and shown in Figure A.1.2(b-d): One half contains the PD while the other half consists of a radio frequency (RF) coplanar waveguide with a ~ 0.5 mm hole in the center to allow x-rays to pass through (i.e., the small hole in Figure A.1.2(c)). The nominal diameter of the x-ray beam at BL 4.0.2 is about 0.1 mm. The LY flux produced by the x-ray-irradiated sample then travels to the photodiode through a ~ 3 mm diameter hole (LY hole in Figure A.1.2(d)) drilled through the sample end piece. All electrical connections between the sample end piece and the vacuum flange are tied around the metal shaft to prevent contact with the walls of the vacuum chamber. The holder is designed such that cryogen can also flow through the holder shaft, allowing it to serve as a cold finger which can bring the sample end piece temperature down to ~ 20 K with liquid helium and ~ 80 K with liquid nitrogen. Moreover, the holder is equipped with a heater which allows for the sample temperature to be controlled during an experiment. For holders capable of measuring an LY signal (i.e., the LY and XFMR holders), the PD terminals are connected to a Bayonet Neill-Concelman (BNC) vacuum feedthrough flange (i.e., electronic ports in Figure A.1.2(a)). Through this component, the in-vacuum PD leads can be connected to a current amplifier residing outside the vacuum system.

A.1.3 – Optimizing Luminescence Yield Detection Sensitivity

The LY signal detection sensitivity of the XFMR instrument was first compared against the LY holder used for conventional static XAS measurement. In these measurements, the BL 4.0.2 XFMR standard sample comprised of (001)-oriented MgO // 12 nm permalloy / 3 nm Cu was measured on both the LY and XFMR holders. All measurements were performed at room temperature (300 K). Figure A.1.3 shows a comparison between LY photocurrents A_{LY} measured for the two different holders, where “PD-LY” correspond with the LY holder while “PD-AXUV” corresponds with the original XFMR holder AXUV20HS1 from Opto Diode Corporation. To

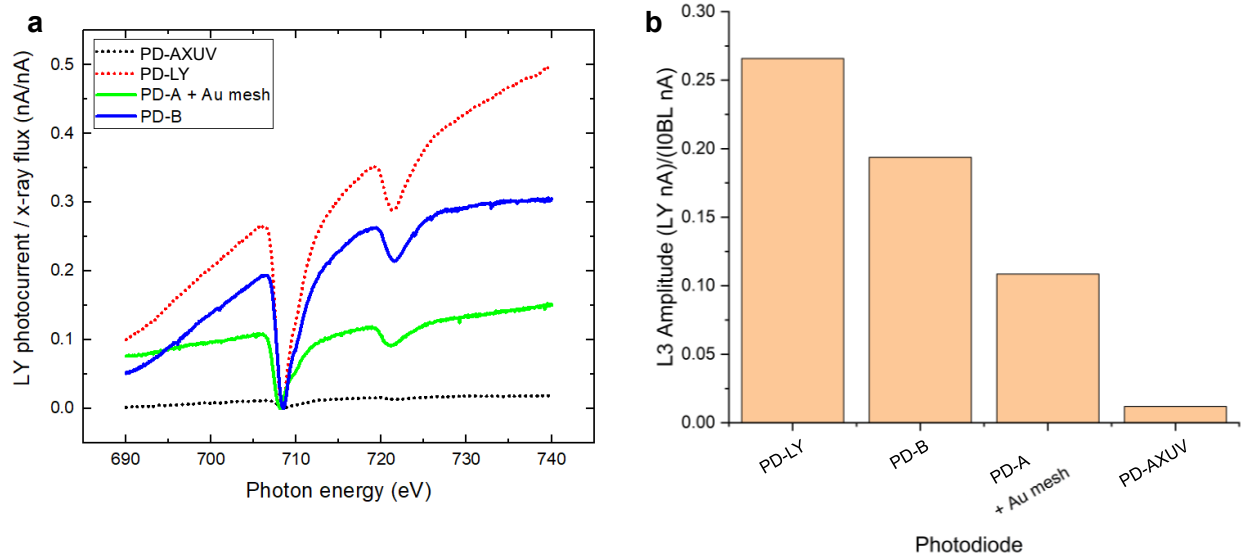


Figure A.1.3: (a) Fe $L_{2,3}$ spectra for a (001)-oriented MgO // 12 nm permalloy / 3 nm Cu film collected using several photodiode configurations. All spectra have been offset so that the L_3 edge minima is set to zero. The LY signal photocurrent was normalized by the flux of the incident x-ray beam. (b) Comparison of the Fe L_3 amplitude measured for several different photodiode configurations. The amplitude is defined by the maximum intensity in (a) at energies below the L_3 edge.

consistently compare the signal strength measured across the different hardware and experiment times, all signals were normalized by the incident x-ray flux photocurrent A_{in} which were measured simultaneously with A_{LY} .

As shown in Fig. A.1.3(a), the LY signal produced by the XFMR holder is substantially weaker in comparison to the signal produced detected by the LY holder. By using the amplitudes of the Fe L_3 XAS signal (i.e., the intensity difference between the edge minimum at ~ 708 eV and the pre-edge maximum at ~ 706 eV in Fig. A.1.3(a)) as a measure of the signal sensitivity, I found that the XFMR holder is approximately 20 times less sensitive to the LY signal changes compared to the LY holder as shown in Fig. A.1.3(b).

This difference in photodetection sensitivities between the two holders can be explained by the different PDs these systems use. The photosensitivity of a PD, also known as responsivity, quantifies how efficiently the diode produces photocurrent when irradiated with different wavelengths of light. The XFMR holder uses an AXUV20HS1 from Opto Diode Corporation (referred to as the “PD-AXUV”) while the LY holder uses an S3588-08 from Hamamatsu Photonics Kabushiki Kaisha (referred to as the “PD-LY”). The responsivities of the PD-AXUV and PD-LY, found in Ref. [143] and Ref. [144] respectively, demonstrate that the former PD is less responsive across the visible light spectrum (400 nm – 700 nm) compared to the latter.

In principle, using the PD-LY in the XFMR holder would resolve any LY signal sensitivity discrepancies between the two instruments. However, several additional constraints must be considered when selecting an optimal diode for XFMR measurements. The first is a geometric constraint associated with utilizing the existing XFMR holder design: the PD must be able to fit within the diode slot (rectangular hole in Figure A.1.2(d)) with dimensions of $\sim 2\text{ mm} \times 5\text{ mm} \times 5\text{ mm}$. The PD-LY is approximately $\sim 30\text{ mm}$ in length and, thus, cannot fit in this holder. Second, the rise time of the PDs must be faster than the rise time of the current amplifier ($30\text{ }\mu\text{s}$) used in the XFMR setup, ideally by at least a factor of 10. Thus, the ideal PD is one which provides the best visible light range responsivity (e.g., the PD-LY), the fastest rise time (e.g., faster than $3\text{ }\mu\text{s}$), and shares the same form factor as the XFMR PD.

Based on a consideration of these factors, two PDs from Hamamatsu Photonics were selected for testing: the S1337-66BR (PD-A) and S1227-66BR (PD-B). The specifications for PD-A and B can be found in Ref. [145] and [146], respectively. Figure A.1.3(a,b) shows the XAS signal strength of the same permalloy reference sample measured using PD-A and B. These results

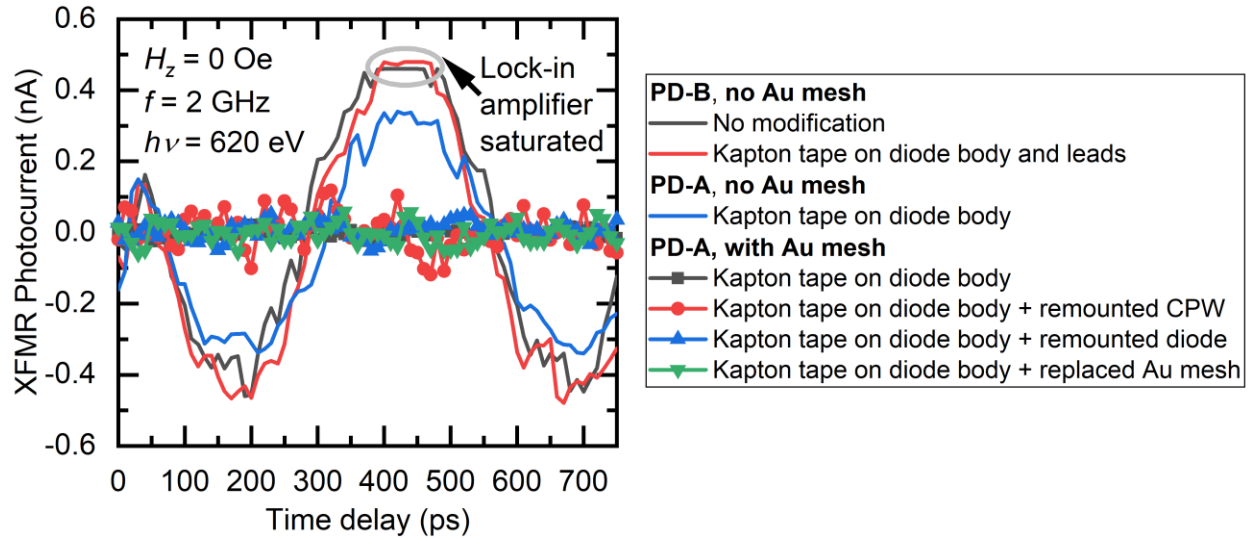


Figure A.1.4: Comparison of XFMR signals captured with a bare (001)-oriented LSAT substrate for different detector hardware configurations. The applied static field, RF frequency, and photon energy are represented with H_{static} , f , and $h\nu$, respectively.

show that using PD-A and B introduces a near order of magnitude improvement in LY detection sensitivity over the PD-AXUV. For the measurement performed with PD-A, the light received by the diode was partially blocked by a gold mesh covering which is rated to block ~40 % of the visible light transmission. The role of this gold mesh will be discussed in the following section.

I note that a complete comparative study between PD-A and B were not performed due to the limited time available to perform diagnostics on the XFMR instrument *with* soft x-rays. The subsequent use of PD-A and a lack of XAS data for PD-A without the gold mesh were a consequence of *ad hoc* diagnostic decisions made during the beamtime (duration on the order of ~24 hours).

A.1.4 – Identification of an XFMR Artifact Signal

While upgrading the PDs resulted in an improvement to the total LY (i.e., static) detection sensitivity, additional tests were conducted to ensure that the ability to perform XFMR (i.e.,

dynamic) measurements were not compromised. For these tests, a bare (001)-oriented LSAT substrate was measured, which is nominally diamagnetic [147]. All measurements were performed at room temperature (300 K). The purpose of using a non-ferromagnetic sample is to test for the potential presence of artifact signals that may have been introduced by the hardware changes. When measuring ferromagnetic samples, a similar diagnostic test may also be conducted by setting the photon energy to two conditions: (1) the energy which maximizes the static XMCD signal (~ 0.2 eV lower than the L_3 resonance edge for Mn in LSMO) and (2) off the elemental resonance. In principle, XFMR delay scans (e.g., Figure A.1.4) performed with condition (2) should produce a flat curve as the XFMR signal is fundamentally an XMCD signal and can only be detected near the elemental resonance.

To test for the presence of an artifact signal, I measured XFMR time delay scans for the LSAT substrate and different detector hardware configurations using an H_{RF} frequency of 2 GHz, $H_{static} = 0$ Oe, and an off-resonance photon energy $h\nu = 620$ eV. The results of these measurements, shown in Figure A.1.4, indicates that using PD-A and PD-B introduces an anomalous sinusoidal signal which shares the same frequency as H_{RF} . Because the presence of this sinusoidal signal cannot reflect the precessional motion of magnetic moments in the sample (i.e., the photon energy is off resonance and no ferromagnetic materials were used), I concluded that this signal is likely an artifact. This artifact signal was never observed when PD-AXUV was installed in the XFMR holder. The most obvious difference between PD-AXUV and PD-A and B is that the latter two diodes are potted with resin while PD-AXUV is uncapped. However, it remains unclear what additional differences exist between the packaging of PD-AXUV and PD-A and B.

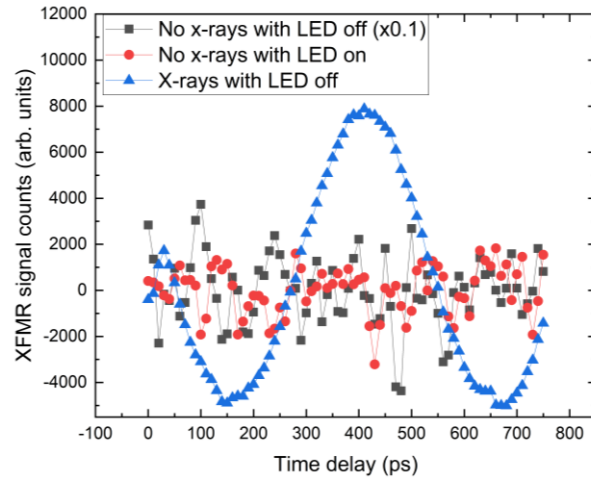


Figure A.1.5: Comparison of XFMR delay scan line shapes in the presence and absence of x-rays and visible range white light illumination from an LED.

A.1.5 – Suppression of the XFMR Artifact Signal

In order to benefit from the LY sensitivity improvements PD-A and PD-B offer over the PD-AXUV, the source of the artifact signal must be identified and suppressed. Assuming that the artifact signal does not reflect a time-varying photon absorption of the sample (i.e., some physical process in the sample), it follows that visible-range photons (i.e., the LY yield signal) are not the source of the artifact signal. Two potential sources of the artifact signal are radio transmission from the coplanar waveguide to the PD or photoelectron emission.

To isolate the source of the artifact signal, several diagnostic tests were performed with PD-A installed in the XFMR end piece. Assuming that radio transmission from the coplanar waveguide was the primary source of the artifact, I performed several XFMR measurements using the same bare (001)-oriented LSAT substrate and three different sample illumination conditions: (1) sample is not illuminated by either x-rays or a white light LED, (2) the sample is only illuminated by a white light LED, and (3) the sample is only illuminated by x-rays. All other

experimental parameters (sample temperature, photon energy, H_{static} , and H_{RF} frequency) are identical to those used in Figure A.1.4.

The results of these measurements are shown in Figure A.1.5, which shows that an artifact signal is only clearly observed when the sample is illuminated by x-rays. The observation that the artifact signal production is correlated with x-ray illumination, but not visible light illumination, suggests that x-ray photoemitted electrons may be the primary carrier of the artifact signal. That is, photoelectrons traveling from the sample to the PD via the LY hole may be facilitating the transfer of an RF signal produced by the coplanar wave guide. In previous XFMR holder designs, the PD packaging and terminals had direct line-of-sight to the interior of the VMM vacuum chamber. It was found that XFMR measurements performed with this end piece design contained substantial noise, which was hypothesized to originate from free photoelectrons impacting the PD. The current XFMR holder is designed such that the PD is encased within a copper body which effectively serves as a Faraday cage to prevent the PD from receiving any photoelectrons. However, the current design does not completely isolate the PD from photoelectrons due to the LY hole, which is a necessary design component to allow the sample-produced LY flux to reach the PD. With these observations, I hypothesized that the artifact signal results from the sample-produced photoelectrons impinging on the PD.

While the reasons behind the true nature of the PD-dependent artifact signal transmission and reception remains unclear, investigation into the matter would not be necessary if the PDs can be shielded from the artifact signal source *without* interfering with the detection of visible light. Investigation into the underlying cause of artifact susceptibility in the PDs would be interesting from a scientific standpoint. But due to the limited availability of beamtime (on the order of a few days granted every year), such an in-depth investigation would be impractical. Thus, the aim of

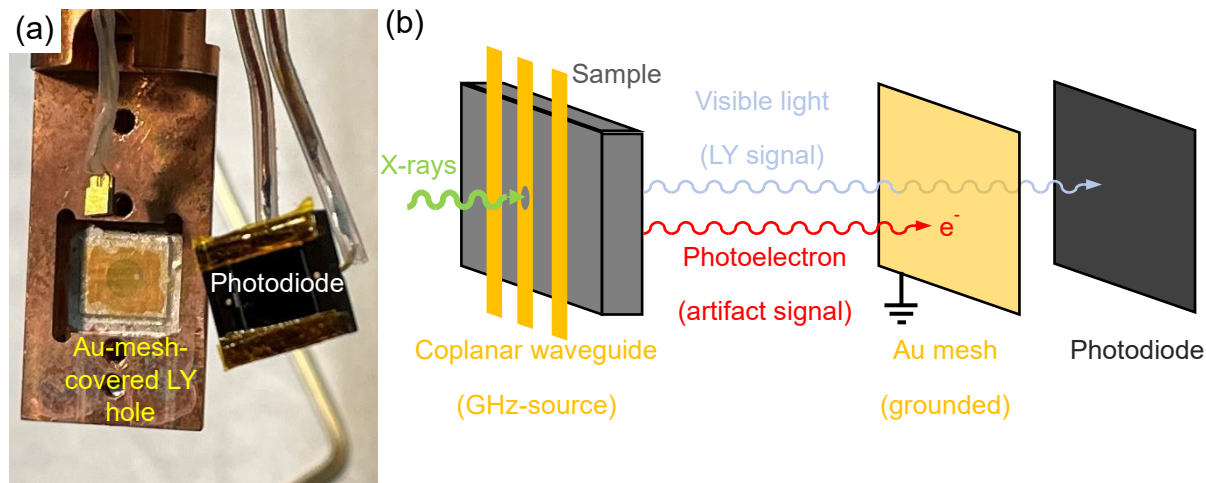


Figure A.1.6: (a) Image of the XFMR end piece with the Au mesh covering the LY hole. (b) Schematic of the hypothesized mechanism behind the XFMR artifact observed in Figures A.1.4 and A.1.5.

this portion of the work is not to understand the full nature of the artifact signal production, but to find a solution to suppress this signal.

In principle, the PD can be completely shielded from photoelectrons if the hole was covered with metal foil that is electrically grounded to the rest of the XFMR holder. However, metal foils are generally opaque and would thus prevent the measurement of an LY signal. The solution I conceived was to cover the hole with a thin Au mesh sheet with $\sim 40\ \mu\text{m}$ hole size. As shown in Fig. A.1.6, my hypothesis was that the mesh wire density was sufficient to block all or enough photoelectrons from reaching the PD to produce a detectable artifact signal while allowing sufficient visible light to pass through to produce a meaningful LY signal. The mesh was mounted directly inside the PD-slot of the XFMR holder and was electrically grounded and mechanically adhered to the Cu body using silver paint. To prevent the PD surface from making full contact with

the Au mesh, two $\sim 1\text{ mm} \times 2\text{ mm} \times 5\text{ mm}$ rectangular pieces of layered Kapton tape were attached to the top and bottom ends of the surface of PD-B.

The off-resonance XFMR measurements performed with the Au mesh on PD-A is shown in Fig. A.1.4 (all curves with symbols), which clearly demonstrate the lack of an artifact signal (i.e., the time-delay scan is flat). Moreover, this observation is reproducible: even after readjusting the end piece hardware and remounting a new Au mesh piece, the measured XFMR scan remains free of the artifact signal. This observation supports the hypothesis that the artifact signal is carried by photoelectrons traveling down the LY hole which are effectively blocked by the Au mesh. Most importantly, the introduction of the mesh still allows PD-A to collect enough light to outperform the old XFMR photodiode as shown in Figure A.1.3. These measurements show that the combination of PD-A with the Au mesh provides a near order of magnitude improvement in LY sensitivity over the original XFMR hardware when measured on the MgO sample.

A.1.6 – Conclusions

In conclusion, I've investigated and implemented methods to improve the measurement sensitivity of the XFMR instrument at BL 4.0.2. Comparing the LY measurement performance between the XFMR and LY sample holders revealed that the XFMR PD was an order of magnitude less sensitive compared to the PD in the LY holder. Through the installation of new PDs with improved responsivity characteristics in the XFMR end piece, the LY signal strengths measured by the XFMR holder became comparable to those produced by the LY holder. While the new PDs enhanced the LY measurement sensitivity of the XFMR setups, an anomalous artifact signal was introduced into time-resolved XFMR measurements. Instrument diagnostics revealed that the potential carrier of this artifact signal were photoelectrons produced by the x-ray-irradiated sample, which were effectively suppressed by the introduction of a Au mesh layer. This Au mesh enabled

the formation of a full Faraday cage around the PD to isolate the PD from photoelectrons, allowing for the suppression of the artifact signal. The instrumentation work performed in this dissertation will improve the measurement capabilities of the XFMR instrument at BL 4.0.2 by allowing for the detection of smaller LY signals, which in turn improves the acquisition of XFMR signals with higher signal-to-noise ratio. Moreover, the ability to suppress the XFMR artifact signal allows for future upgrades of the photodiode without concerns for potential diode-dependent artifact susceptibility.

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